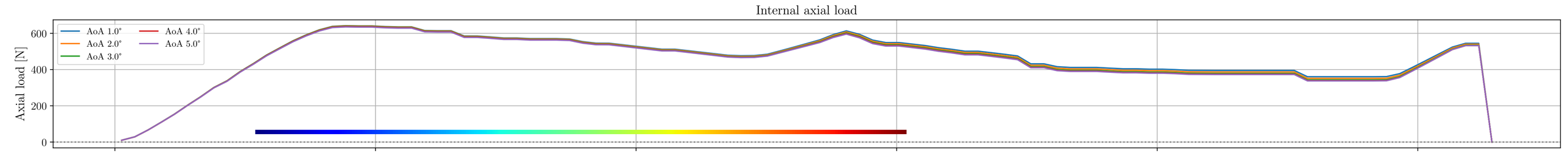
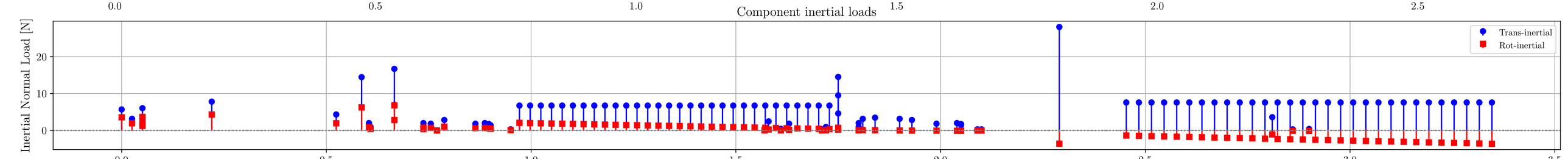
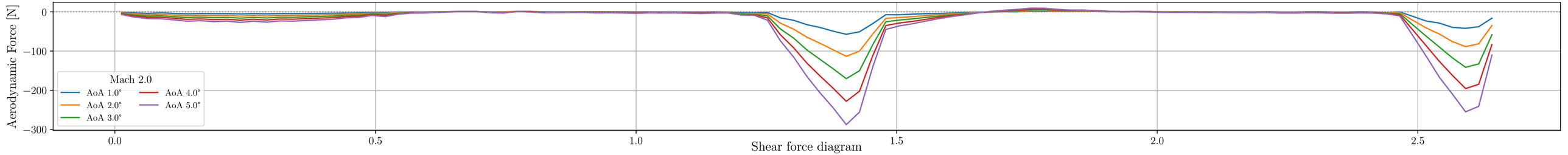
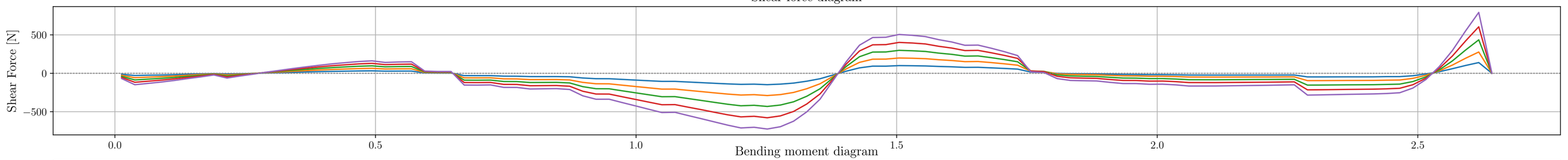
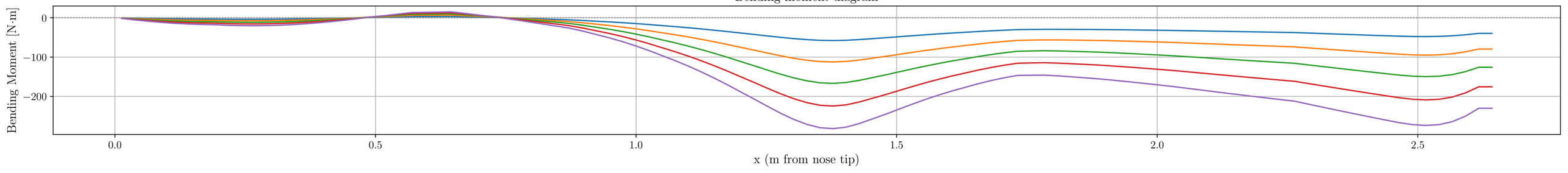

50000
100000
150000
200000
250000
300000
p [Pa]


Internal axial load
AoA 1.0°
AoA 2.0°
AoA 3.0°
AoA 4.0°
AoA 5.0°
Axial load [N]
0
200
400
600
0.0
0.5
1.0
1.5
2.0
2.5


Component inertial loads
Trans-inertial
Rot-inertial
Inertial Normal Load [N]
0
10
20
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Aerodynamic normal force


Mach 2.0
AoA 1.0°
AoA 2.0°
AoA 3.0°
AoA 4.0°
AoA 5.0°
Aerodynamic Force [N]
0
−100
−200
−300
0.0
0.5
1.0
1.5
2.0
2.5
Shear force diagram


Shear Force [N]
500
0
−500
0.0
0.5
1.0
1.5
2.0
2.5
Bending moment diagram


Bending Moment [N·m]
0
−100
−200
0.0
0.5
1.0
1.5
2.0
2.5
x (m from nose tip)